Netflix Data Analysis

Netflix is a subscription-based streaming service that allows members to watch TV shows and movies without commercials on an Internet-connected device.

1. Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objs as go
import plotly.offline as pyoff
import plotly.figure_factory as ff
import cufflinks as cf
import calendar
import warnings
from collections import Counter
from wordcloud import WordCloud, STOPWORDS
from PIL import Image
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
from sklearn.linear_segment import LinearSegmentationModel
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
from sklearn.metrics import mean_squared_log_error
from sklearn.model_selection import cross_validation
from sklearn.model_selection import KFold
from sklearn.model_selection import GridSearchCV
from sklearn.model_selection import RandomizedSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import Normalizer
from sklearn.preprocessing import RobustScaler
from sklearn.preprocessing import PolynomialFeatures
from sklearn.decomposition import PCA
from sklearn.decomposition import TruncatedSVD
from sklearn.decomposition import LatentDirichletAllocation
from sklearn.decomposition import NMF
from sklearn.cluster import KMeans
from sklearn.cluster import MiniBatchKMeans
from sklearn.cluster import AgglomerativeClustering
from sklearn.cluster import DBSCAN
from sklearn.cluster import MeanShift
from sklearn.cluster import SpectralClustering
from sklearn.cluster import Birch
from sklearn.cluster import AffinityPropagation
from sklearn.cluster import FeatureAgglomeration
from sklearn.cluster import OPTICS
from sklearn.cluster import SpectralBiclustering
from sklearn.cluster import SpectralCoclustering
from sklearn.cluster import VectorQuantizer
from sklearn.cluster import MiniBatchNMF
from sklearn.cluster import LocalOutlierFactor
from sklearn.cluster import IsolationForest
from sklearn.cluster import OneClassSVM
from sklearn.cluster import EllipticEnvelope
from sklearn.cluster import LabelPropagation
from sklearn.cluster import LabelSpreading
from sklearn.cluster import BaseEstimator
from sklearn.cluster import ClusterMixin
from sklearn.cluster import TransformerMixin
from sklearn.cluster import BaseSpectral
from sklearn.cluster import SpectralMixin
from sklearn.cluster import BaseDecomposition
from sklearn.cluster import DecompositionMixin
from sklearn.cluster import BaseNeighbors
from sklearn.cluster import NeighborsMixin
from sklearn.cluster import BaseSupervised
from sklearn.cluster import SupervisedMixin
from sklearn.cluster import BaseUnsupervised
from sklearn.cluster import UnsupervisedMixin
from sklearn.cluster import BaseMultivariate
from sklearn.cluster import MultivariateMixin
from sklearn.cluster import BaseMulticlass
from sklearn.cluster import MulticlassMixin
from sklearn.cluster import BaseMultilabel
from sklearn.cluster import MultilabelMixin
from sklearn.cluster import BaseMultiobjective
from sklearn.cluster import MultiobjectiveMixin
from sklearn.cluster import BaseMultivariate
from sklearn.cluster import MultivariateMixin
from sklearn.cluster import BaseMulticlass
from sklearn.cluster import MulticlassMixin
from sklearn.cluster import BaseMultilabel
from sklearn.cluster import MultilabelMixin
from sklearn.cluster import BaseMultiobjective
from sklearn.cluster import MultiobjectiveMixin
```

2. Import Dataset

```
df = pd.read_csv("netflix_titles.csv")
```

3. Exploratory & Descriptive

```
df.shape
```

```
(8807, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtype: int64, object
memory usage: 1.1+ MB
```

```
df.describe()
```

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

```
df.describe(include='object')
```

	show_id	type	title	director	cast	country	date_added	rating	duration	listed_in	description
count	8807	8807	8807	6173	7982	7976	8797	8803	8804	8807	8807
unique	8807	2	8807	4528	7692	748	1767	17	220	514	8775
top	s1	Movie	Dick Johnson Is Kidding	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	TV-MA	1 Season	Documentaries	Documentary about the life of Richard Dawkins.
freq	1	6131	1	19	19	2818	118	3207	1793	334	1

```
df.describe(include='int64')
```

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

```
df.describe(include='float64')
```

```
NotImplemented
```

```
df.nunique()
```

```
show_id       8807
type             2
title         8807
director      4528
cast          7692
country        748
date_added    1767
release_year    74
rating          17
duration       220
listed_in      514
description   8775
dtype: int64
```

```
df.isnull()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	False	False	False	True	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False